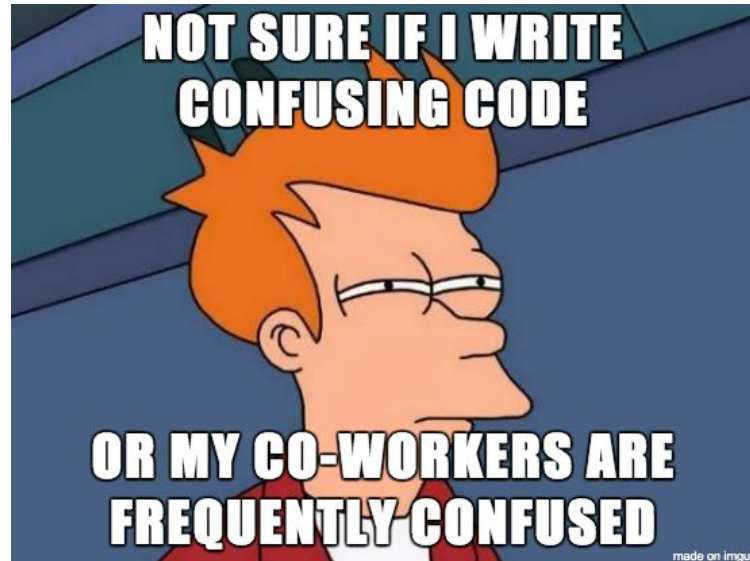


Code quality is misunderstood
... and it's barely taught anywhere 🥲



I'm Alex

- My goal: **Teach you to write clean, high-quality code** 🧽
- Previously a Tech Lead at Course Hero, Meta, and Robinhood
- Course Hero Android app had the highest NPS during my tenure
- Top 5% code committer and reviewer at Meta



I ❤️ Shipping Quality Software



Random Name Picker - 1,000,000+ Users,
4.7 stars out of 5 after 5,600+ reviews



Random Number Generator - 680,000+
Users, 4.8 stars out of 5 after 6,500+ reviews



SQL Practice Pro - 550,000+ Users,
4.9 stars out of 5 after 9,300+ reviews

So many skills to learn...

Experimentation Architecture SQL Agile
Blockchain IDE Tricks Shell Scripting
CI/CD Project Management
Meeting Management OOP Microservices Debugging
Distributed Systems



Code Quality

Objectives

-  Understand what **good code looks like** (and *doesn't look like*)
-  Set up the right **growth mentality** around code quality
-  Learn how to submit a **quality pull request**
-  Avoid common **messy code patterns**



Low Performing Engineers



- Messy code
- Poor communication
- Slow debugger



- Doesn't take feedback well
- Reactive instead of proactive
- Messy code



- Repeats mistakes often
- Messy code
- Procrastinates

High Performing Engineers



- High code quality
- Pays attention to detail
- Master bug squasher



- Humble and treats all feedback as a gift
- High code quality
- Expert mentor and leader



- Always thinking 10 steps ahead
- Deep product sense and user empathy
- High code quality

Good Code

- Proactive thinking
- Attention to detail
- Respect for teammates
- Empathy for users
- Clear communication



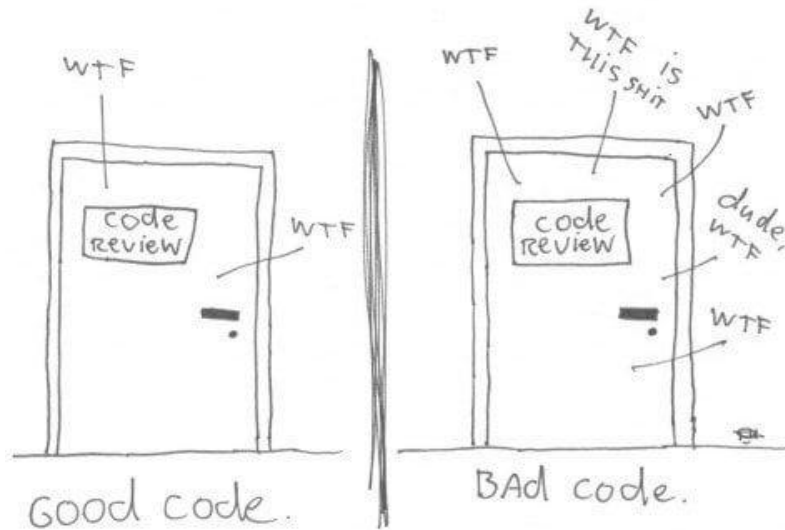
Bad Code

- Reactive mindset
- Laziness
- Disrespect for peers
- Callousness towards users
- Messy communication



Code quality > Code velocity

The ONLY valid measurement
of code quality: WTFs/minute



(c) 2008 Focus Shift

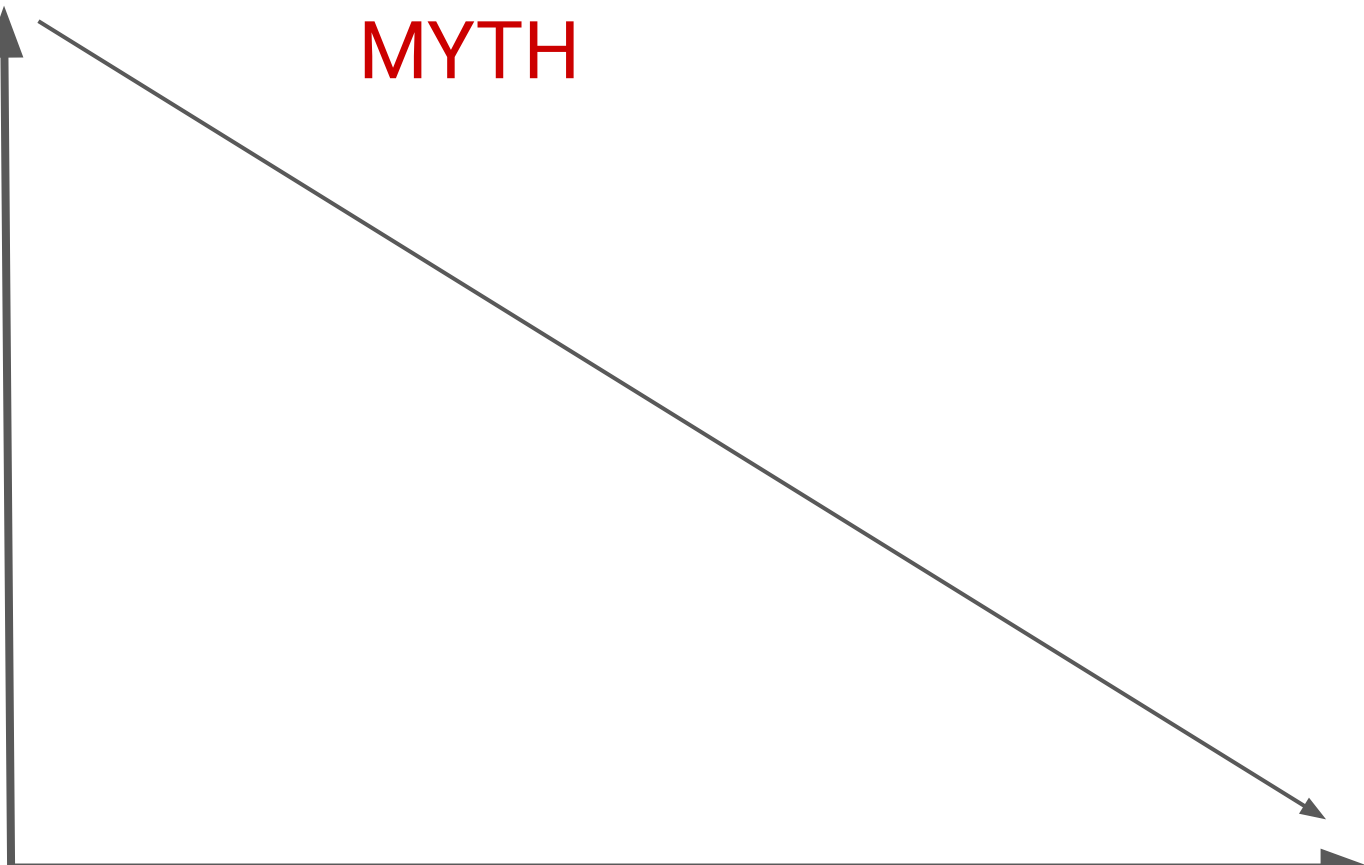
Quality And Velocity Are Best Friends



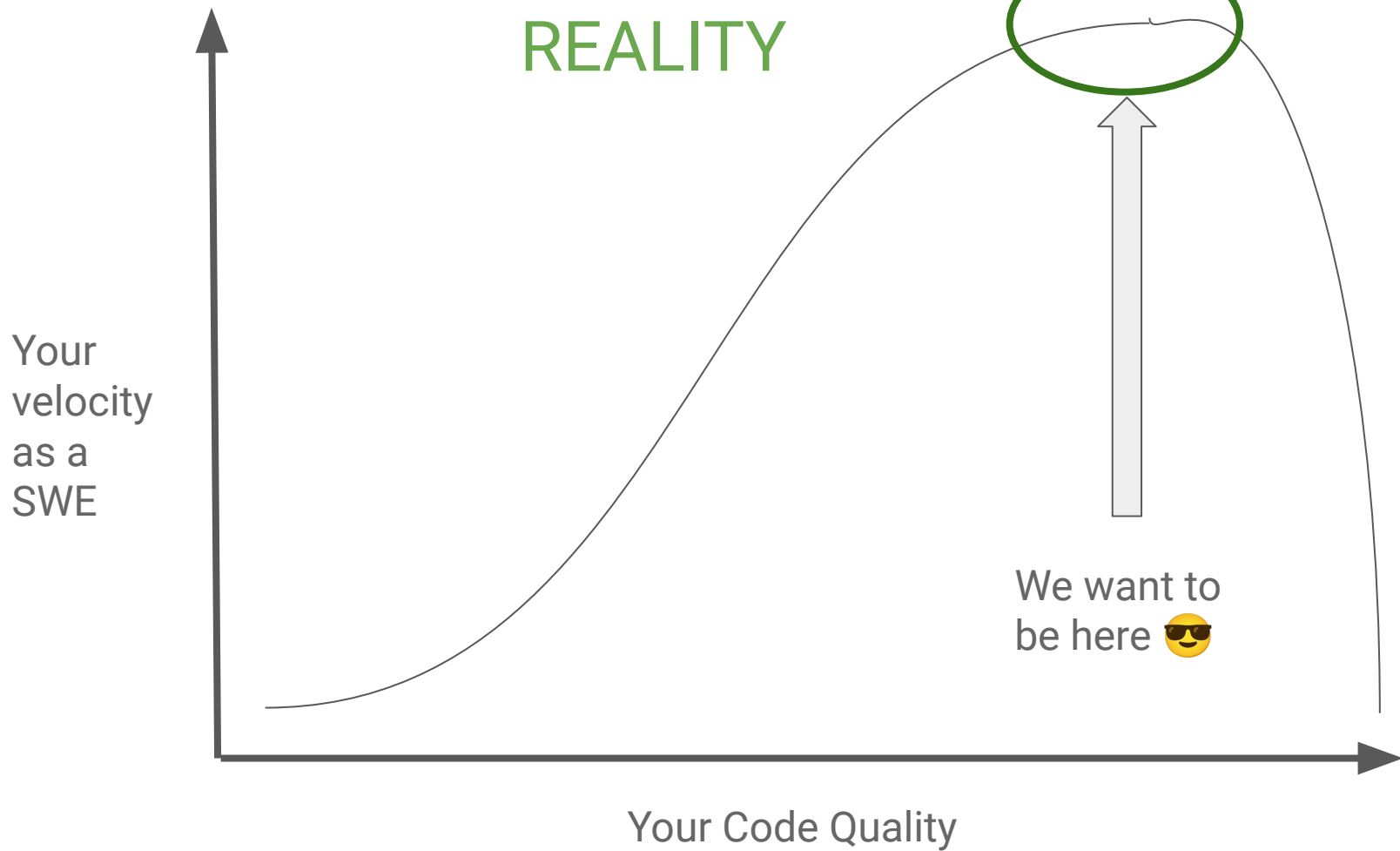
MYTH

Your
velocity
as a
SWE

Your Code Quality



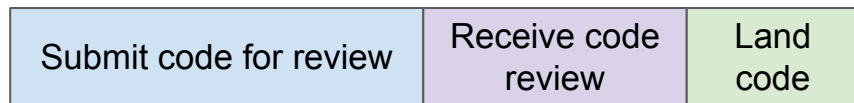
The diagram is a 2D coordinate system with a vertical y-axis and a horizontal x-axis, both ending in arrowheads. A straight line with a negative slope connects a point on the y-axis to a point on the x-axis. The word 'MYTH' is written in red, bold, uppercase letters in the upper-middle area of the graph. The y-axis is labeled 'Your velocity as a SWE' and the x-axis is labeled 'Your Code Quality'.



n00b engineer



Pro engineer



Having The Proper Mentality

What makes code “good”?

A word cloud of programming concepts and practices. The words are arranged in a circular pattern around the center. The words include: Design patterns, Formatting, MVC, NoSQL, Comments, Speed, Modularization, ML Integration, OOP, Small Files, Blockchain, 90% unit test coverage, Spaces Instead Of Tabs, and Factory Pattern.

Design patterns

Formatting

MVC

NoSQL

Comments

Speed

Modularization

ML Integration

OOP

Small Files

Blockchain

90% unit test coverage

Spaces Instead Of Tabs

Factory Pattern

Product Quality

Remember: End-Users Don't Care

- They just want a good user experience (UX)
 - They have no idea how “clean” your code is
 - Most don't even fully understand what code is
- Always put the UX first
 - You'll be surprised at how often “clean” code can have UX negatives



	Bad UX	Good UX
Bad Code		
“Good” Code		

All Roads Lead To The End-User

Scalability



Add features faster

Debuggability



Squash user bugs

Modularization



Testability

Speed



Lower user frustration

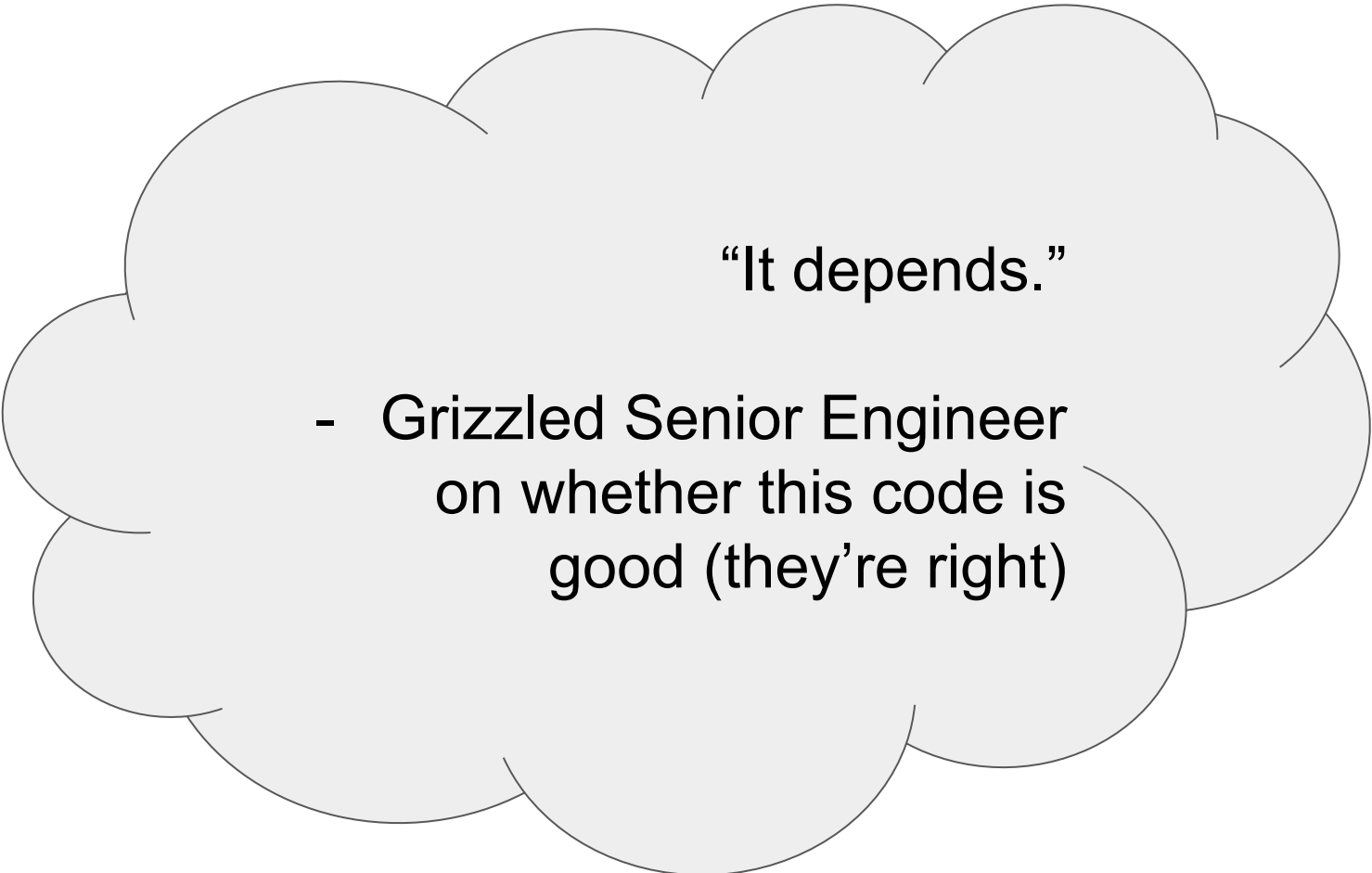
Readability



Smooth onboarding



How does this serve the end-user?



“It depends.”

- Grizzled Senior Engineer
on whether this code is
good (they're right)

There Are No Shortcuts

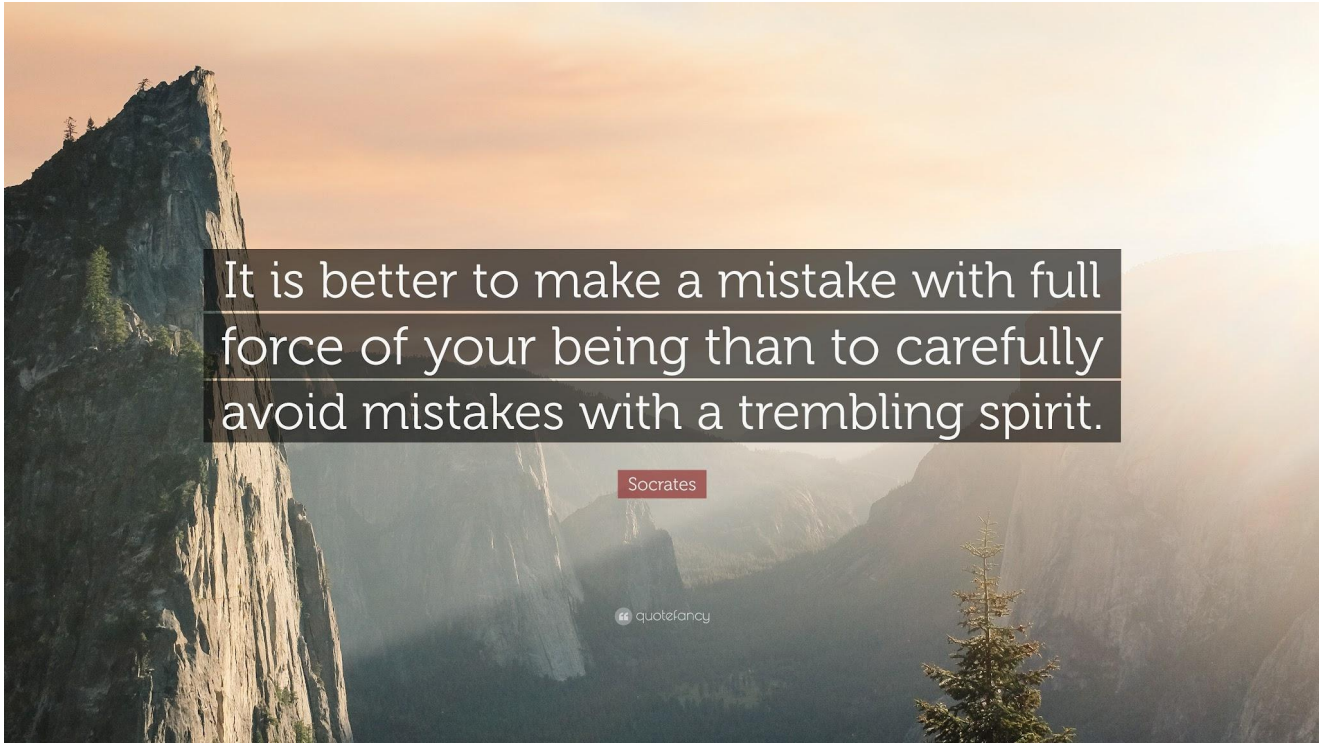
- There is no such thing as objectively “clean code”
- Code quality and software engineering as a whole is all about **context**
- It all depends on your situation
 - Code that is good in one codebase can be horrible in another
- Push to understand trade-offs



These DON'T Automatically Improve Code Quality

- ✗ Using the most modern and “cutting edge” frameworks
- ✗ Plugging in the most scalable solution
- ✗ Copying the tech stack of your favorite FAANG company
- ✗ Making the code more compact
- ✗ Integrating the latest technology buzzword

Embrace The Pain

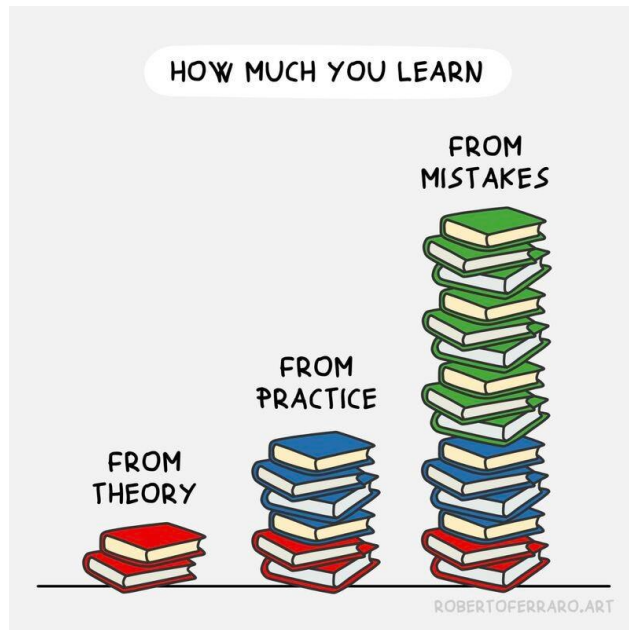
A dramatic landscape photograph of a mountain peak, likely El Capitan in Yosemite National Park, at sunset or sunrise. The sky is a warm, hazy orange and yellow. The mountain face is dark and rugged, with some greenery visible on the lower slopes. A quote is overlaid in the center of the image.

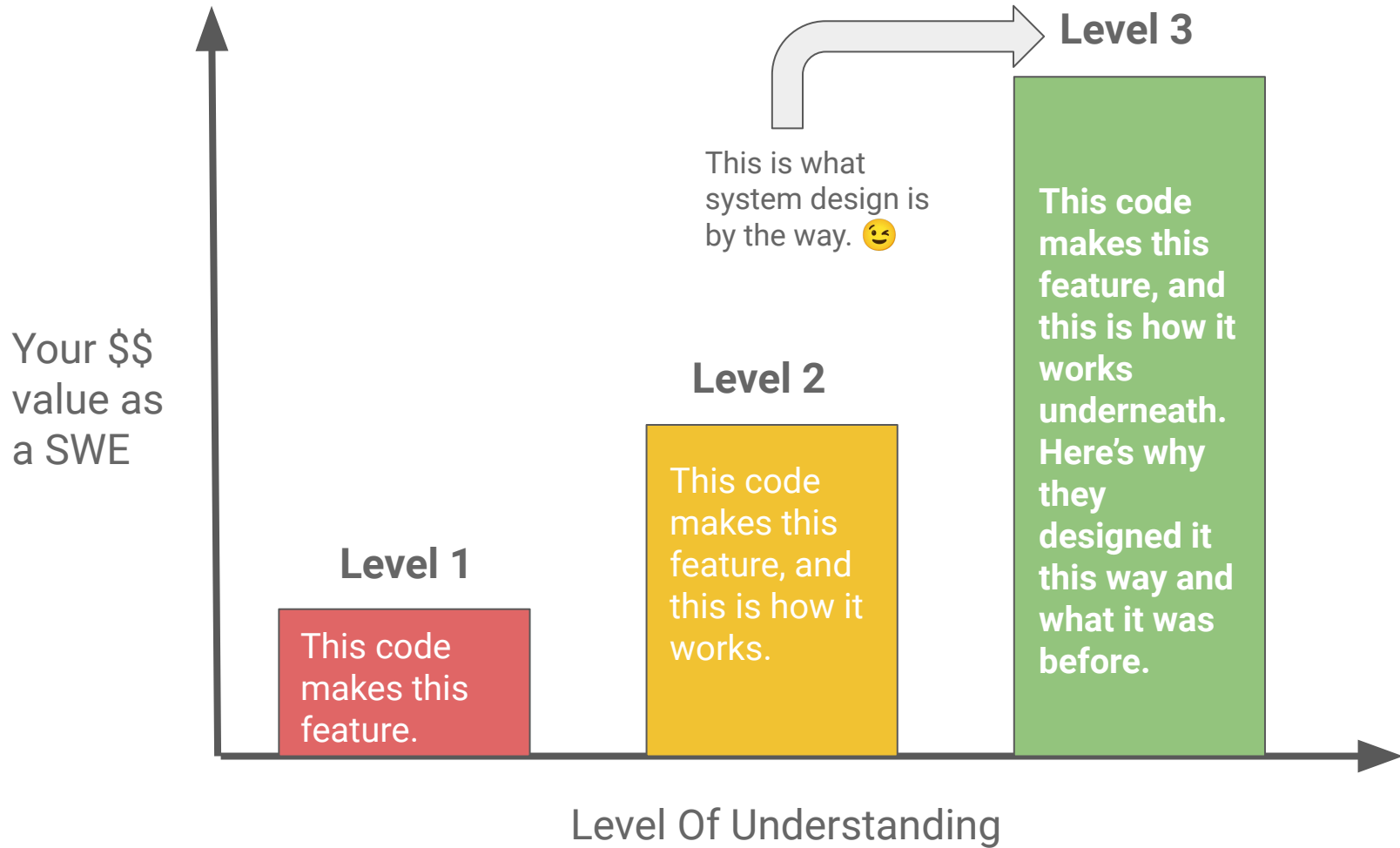
It is better to make a mistake with full
force of your being than to carefully
avoid mistakes with a trembling spirit.

Socrates

quote fancy

- The best way to learn how to write good code is to write bad code first
- It's about the journey, not the destination
- It's an opportunity with your code when:
 - It breaks
 - It's a pain to extend
 - It can't be read by your teammates

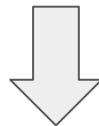




You render a list of items with a ListView.



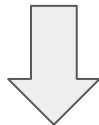
It starts getting laggy when you have 100+ items in it. 😞



You realize that RecyclerView does what you just did. 💡



You implement view recycling manually in your ListView. 💪



You look out for view recycling in iOS and web development. 🚀

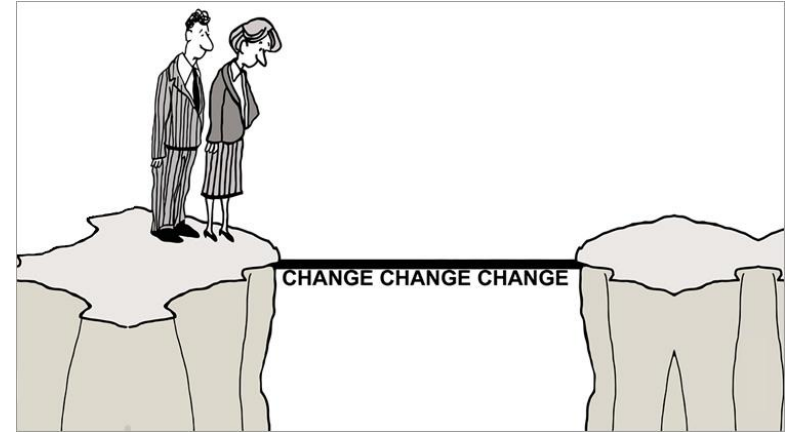


Code Quality Isn't Static



The Right Code For The Job Constantly Changes

- You can't just write "good code" once and never touch it again
- Constantly adapt to the changing landscape - *This is hard work*
- *What got you here won't get you there*
 - **What got others there won't help you here**



- Monolith
- No automated test coverage
- No backups
- Unencrypted traffic
- No rate limiting

10,000,000 users

“Are you insane?” 😬

- Microservices
- 90%+ automated test coverage
- Several layers of backups
- Multi-module builds with artifact caching
- End-to-end encryption across all features
- DDoS protection

1,000 users

“Are you insane?”





Code Quality Case Study: Instagram & Robinhood





Instagram

- **Userbase:** Over 1 billion with most users in developing markets.
India is the most represented country.
- **Code style:**
 - Very vanilla - No external libraries
 - Verbose but hyper-optimized
- **Why:** Binary size and performance are top priorities.



- **Userbase:** ~10 million users, >99% in the United States with high phone quality. Startup willing to take risks.
- **Code style:**
 - Very modern with cutting-edge libraries
 - Relentlessly concise
- **Why:** Leverage user demographic for a more ambitious architecture. Push the technological boundary.

Better Code Strategies

It's All About Focus

“I fear not the man who has practiced 10,000 kicks once, but I fear the man who has practiced one kick 10,000 times.”

Bruce Lee

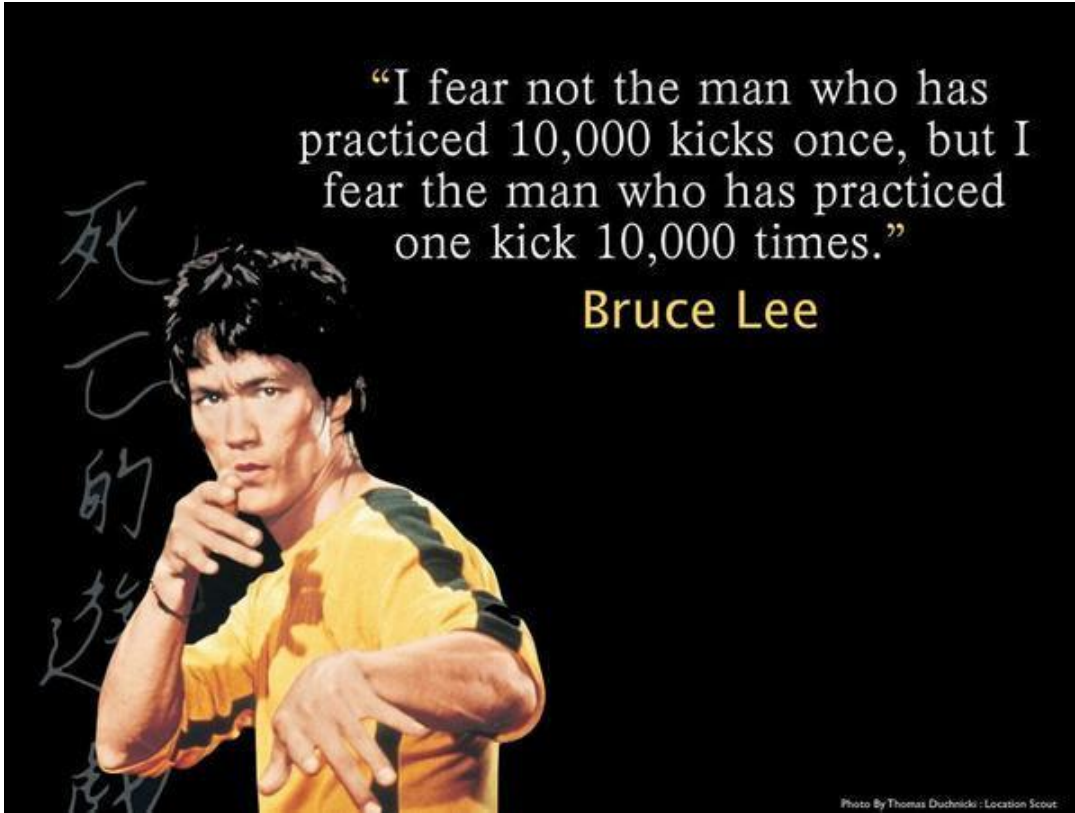


Photo By Thomas Duchnicki - Location Scout

Become The Top 10%

- Start off your career by picking 1 tech stack and mastering it
- How to tell if you have become a master?
 - Everyone starts asking you for guidance on it
 - You become a thought leader
- Avoid “full-stack” early on
 - Lean into one



Who's More Hireable?

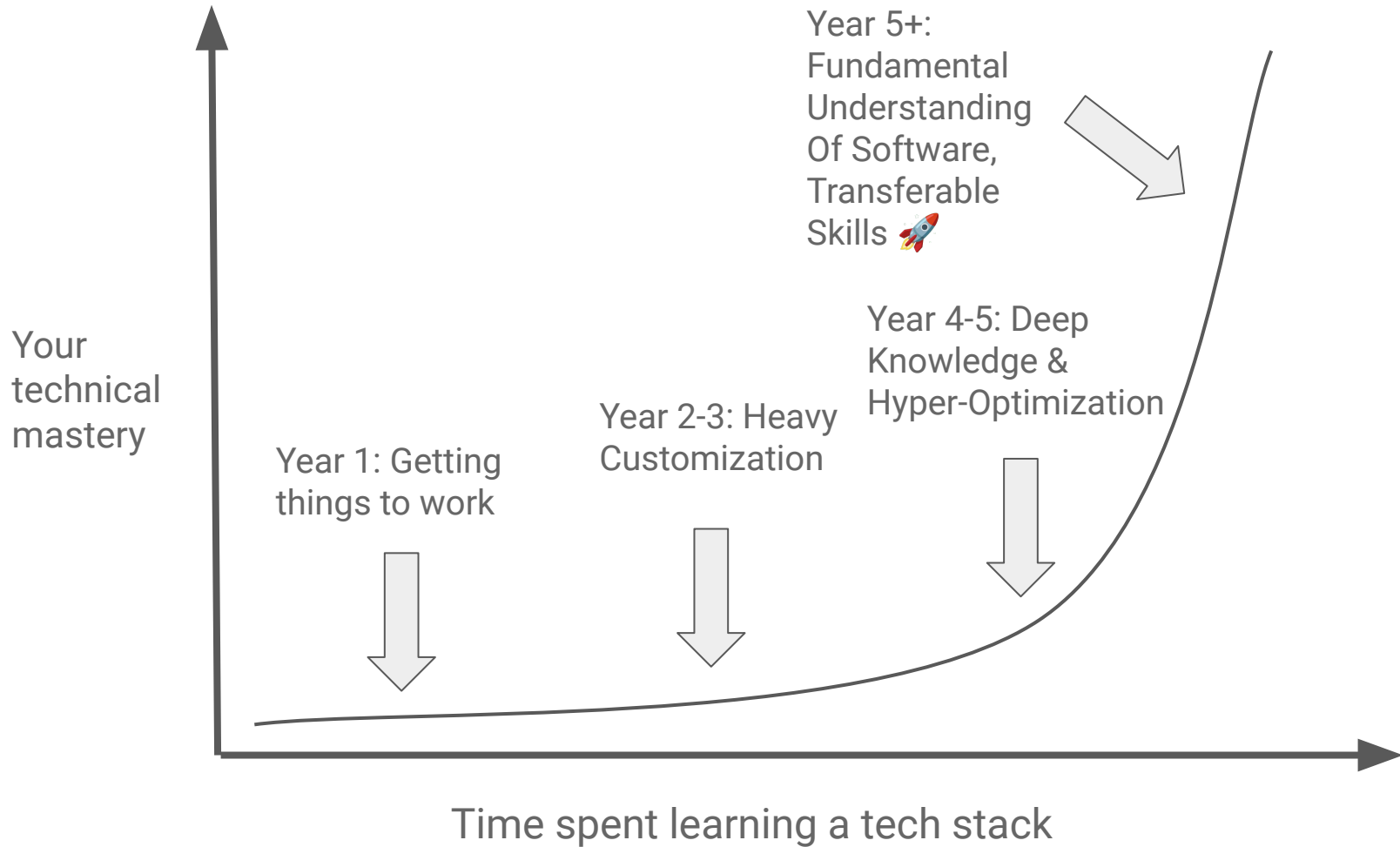


- 4/10 in backend development
- 5/10 in frontend development



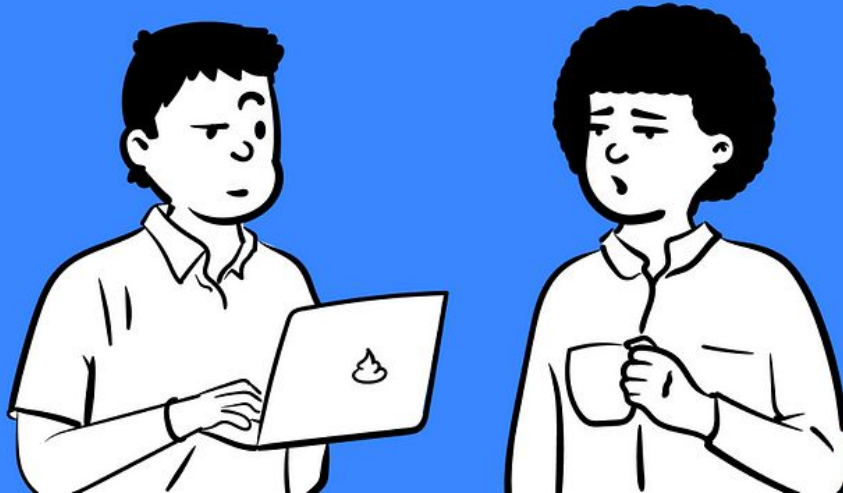
- 9/10 in frontend development





Sweat The Details

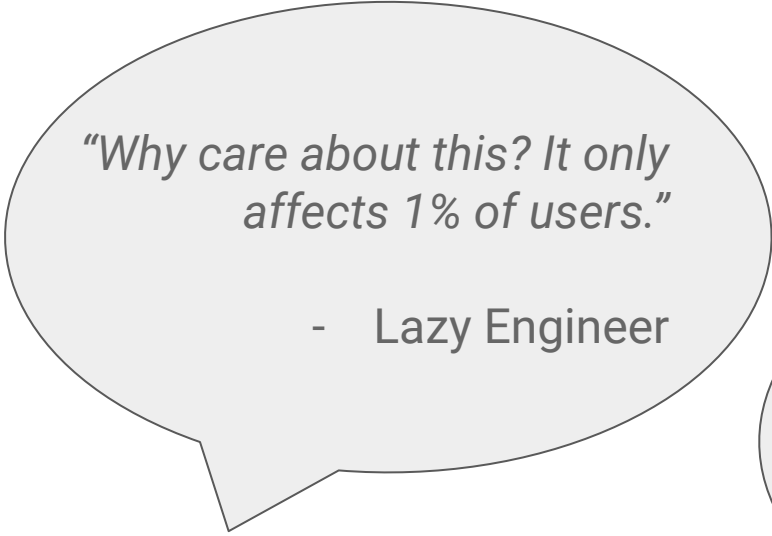
Are you aware of this edge case?



Call Out Edge Cases

- Software is a space where mistakes really hurt
- Being detail-oriented makes you incredibly valuable
 - Especially in Big Tech
- Push your software to the limits in all phases
 - Planning
 - Coding
 - Dogfooding





"Why care about this? It only affects 1% of users."

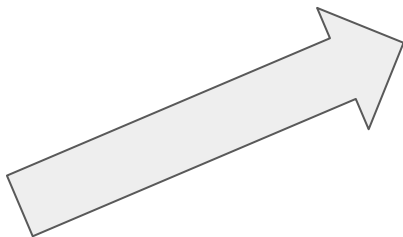
- Lazy Engineer



"You realize that 1% of 1 billion users is 10 million people right?"

- Reasonable Engineer

“So how do I figure out what the edge cases are?”



Assume your users are like this 😂

Don't Assume That...

- Your users understand how to use your product 🙄
- Your users won't make mistakes 🤔
- Your users aren't trying to hurt you 😡
- Your users have a nice computer or phone like you do 💻
- Your users have a steady internet connection 📶
- Your users read your onboarding UX 🤦

**The defining trait of a bad engineer is
that they run away from hard problems.**

API call fails

Rare browser

Invalid

Blank
email

email

Mobile viewport

Blank password

Adblock
DDoS Attack

Landscape
view

Password isn't secure
enough

Form goes
off-screen

Email already taken

Email harvesting

Passwords
don't match

Password too long

Old browser

API call

User double clicks
button

Dark mode

is slow

Email

email@email.com

Choose password

Confirm password

SUBMIT



"Why do these companies have thousands of engineers? They're mostly just CRUD applications."

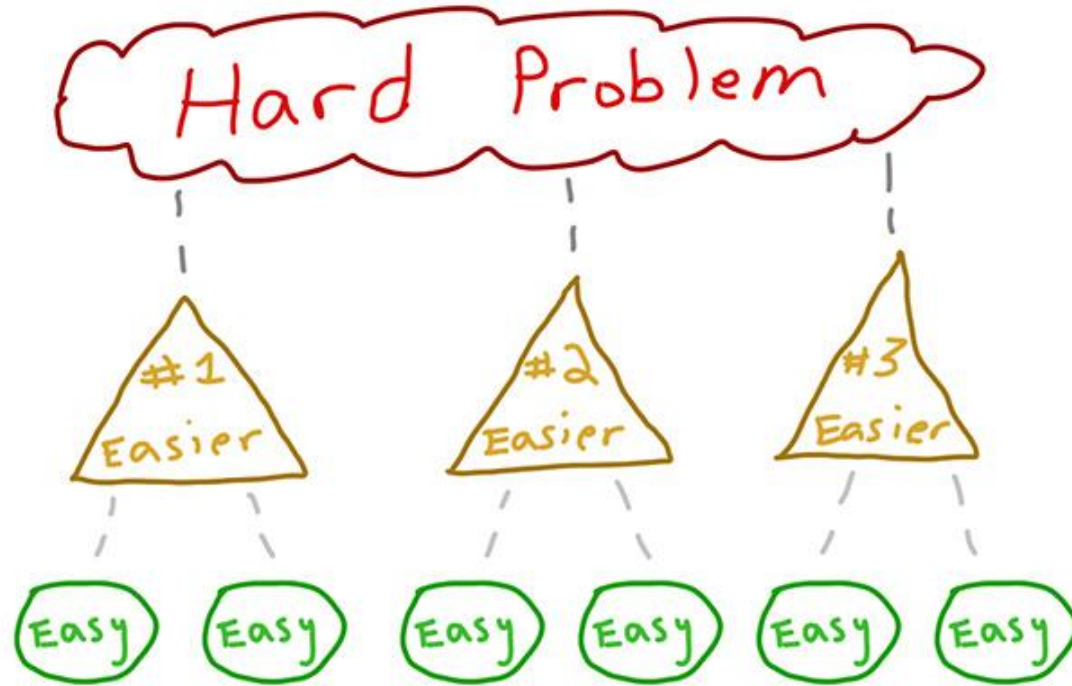


****Watches Taro's sign-up form case study about edge-cases.****

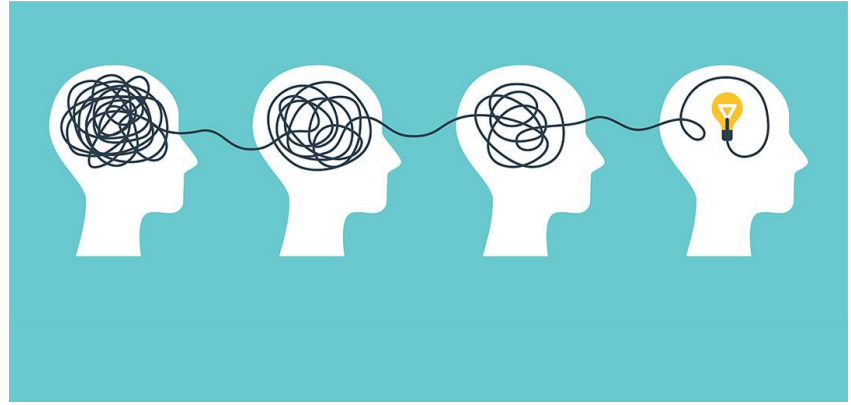


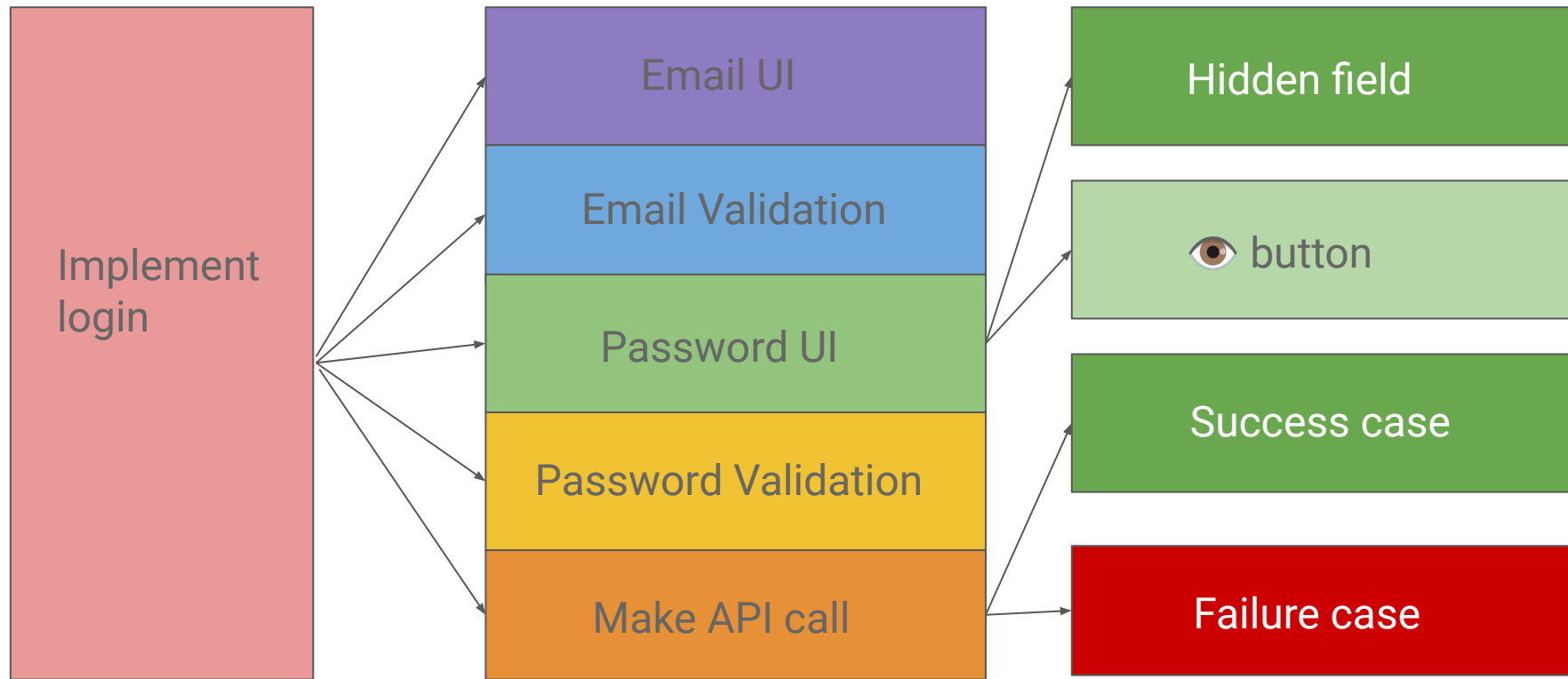
"Ah yep, I get it now."

Break It Down



- Split the problem into small chunks that aren't scary
 - Split further down if necessary
- Each chunk = 1 pull request
- Decomposition is a vital skill everywhere in tech:
 - Debugging
 - Delegation and technical leadership
 - System design





Lay Out The Plan



I got a plan.

- Use your task/design doc to lay out the plan
 - <2 weeks: Use task
 - >2 weeks: Use a doc
- Put in all the ingredients from before
 - Decomposed chunks (rough diff stack)
 - Edge cases
 - Overall strategy (2 approaches if possible)
- Socialize with team



✓ Completed



[Web] Allow users to comment on questions anonymously



Helpful Tarodactyl • Taro Community

a few seconds ago

Share your thoughts to support another member of the Taro community!



☒ Comment anonymously

Comment

Requirements

We'll need to:

1. Add a "Comment anonymously" checkbox on the composer that is unchecked by default
2. Add a "isAnonymous" flag on the Q&A thread item
3. Disable profile clicks if the comment was posted anonymously
4. Make sure that anonymously posted comments don't show up on a user's profile
5. Have the toggle generate an identity randomly, covered in-depth in the section below.

- Overall strategy (data model and UI/UX) 
- Edge cases 

Identity

If we're doing dinosaur-themed:

- Helpful Tarodactyl
- Supportive Tarodactyl
- Collaborative Tarodactyl
- Thoughtful Tarodactyl
- Friendly Tarodactyl

If we're sticking with generic human-ey titles:

- Helpful Taro Member
- Supportive Taro Member
- Collaborative Taro Member
- Thoughtful Taro Member
- Friendly Taro Member

Providing multiple approaches



Subtitle will always be "Taro Community".

Edge Cases

1. **Identity leaks on mobile** - Mobile doesn't support profile, so as long as you're burning in the identity properly, no mobile changes are required.
2. **User is commenting on their own question** - In this case, disable anonymous commenting as asking a question is already anonymous.

For v1, don't support editing a response to anonymous or vice-versa. That's going to make the code mega-spaghetti.

Bonuses

1. Add a little dice button that RNGs the identity from #1
2. AI generate the image on the fly
3. Add more supported images and titles

- Thorough edge case coverage 🤔
- Forward thinking 

How To Handle Migrations



Refactors Should Be Global

- Codebase *consistency* is crucial for understanding
- Given these 2 codebases:
 - 50% “good” pattern, 50% legacy pattern
 - **100% legacy pattern** 👑
- *“I would rather work on a codebase that’s consistently terrible than one that’s half trying to be good.”*



Remember: This Is How New Engineers Code

1. Get a task 
2. Find existing code that accomplishes something >70% similar to that task 
3. Copy paste that code 
4. Tweak it to fit your task's specific parameters 
5. Try to ship it 

“Modern”
pattern
component

“Modern”
pattern
component

Legacy
pattern
component

Legacy
pattern
component

“Modern”
pattern
component

“Modern”
pattern
component

Legacy
pattern
component

Legacy
pattern
component

“Modern”
pattern
component

Legacy
pattern
component

New Engineer



“wtf”

Make It A Little Better Each Time

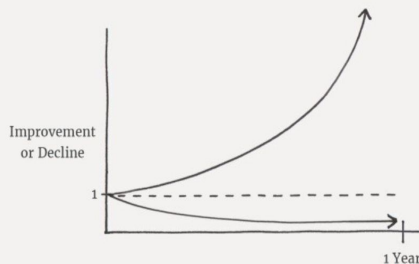


- Pay attention to pain points as you go through code
 - Especially recycled code
- Example questions to ask:
 - Can you do the same with less code?
 - Can you make it more readable?
 - Are edge cases missed?
 - Can you make it faster?

The Power of Tiny Gains

1% better every day $1.01^{365} = 37.78$

1% worse every day $0.99^{365} = 0.03$



“There is only one way to eat an elephant: a bite at a time. 🐘”

Unused code

Unclear variable
+ method names ✨

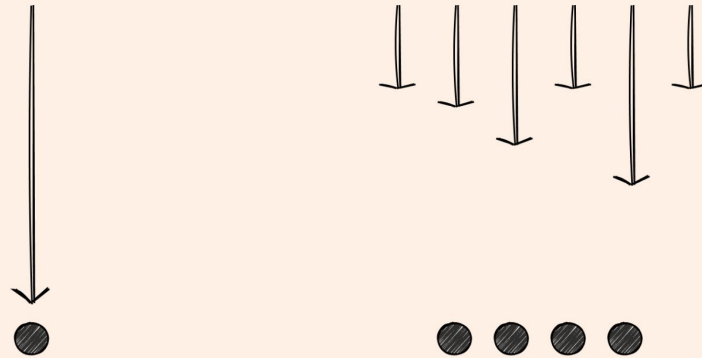
Verbose 😇
legacy
APIs

Monolithic God
Class

Pull Requests

One Diff, One Thesis

Success = Focusing on 'One Thing' at a time



- Keep your diffs atomic and focused
 - < 250 SLOC is a generally good size
- Makes life easier for everyone:
 - Smooth to write as the author
 - Quick to review
 - Easy to land and feel good about progress 🙌



1 Diff, 1 Thesis 🧐

- <3 days to write
- Easy to test and prove that the code works
- Context section is 1 paragraph or less
- <250 SLOC



1 Diff, Too Many Theses 😞

- 1 week+ of pure coding
- Takes forever to test all cases and show that everything works
- Context section is an essay
- 500+ SLOC



Add all login
functionality

Email UI

Email Validation

Password UI

Password Validation

Make API call

Share The Context



- **Goal:** Explain why your diff exists and why it looks the way it does
- **Ingredients:**
 - Business value added (link the connected task)
 - Overall approach explanation
 - Alternate approaches
 - Relevant history
 - *Self-comment tricky areas*

Feature/youtube-full-screen #19

🔒 Closed

prnvptl wants to merge 4 commits into `Gear61:master` from `prnvptl:feature/youtube-full-screen` 

Conversation 1

Commits 4

Checks 0

Files changed 5

 prnvptl commented on Feb 18, 2022 • edited by Gear61 ▾ ...

Closes [#18](#)

Enables Youtube video full screen with a clean UX

- User will be able to go to full screen and back
- Video will NOT reset or go out of sync with this interaction
- No need to go outside of the app for the lesson at all :^)
- Uses Kotlin and cleans up the watch content fragment
- Uses out of the box WebView with a custom chrome client

Links the relevant issue,
outlines goal of PR

Lays out overall approach

Proactively talks about
polish and edge-cases

Conversation

1

Commits

4

Checks

0

Files changed

Commits on Feb 17, 2022

Enables Youtube/web full screen with a clean UX



prnvptl committed on Feb 17, 2022

Commits on Feb 18, 2022

Simplifying full screen webview logic



prnvptl committed on Feb 18, 2022

Convert VideoEnabledWebChromeClient to Kotlin



prnvptl committed on Feb 18, 2022

Using safe access all possible places



prnvptl committed on Feb 18, 2022

Clear, readable commits



After trying many solutions, I managed to stitch this one together which uses a custom chrome web client and overrides onShowCustomView and onHideCustomView which in turn, implements hiding and showing a full-screen video.

Refactored some code in the `WatchContentFragment`.

Alternate solutions:

1. [Youtube Video Player API](#): costs \$ after 10K queries / day. implementation is possible but cumbersome and will require more effort

2. [android-youtube-player](#)

Allows for ALOT of customizability, however seems like overkill for now, and would also require

much more effort to even get a simple web view working like the current web view. A simple

solution of showing full screen would just be to show a button outside of the web view. not the best UX.)

Sources:

- <https://stackoverflow.com/questions/15768837/playing-html5-video-on-fullscreen-in-android-webview/16179544#16179544>

Explains technical strategy

Describes pros/cons of

alternate approaches and why they weren't chosen

Links relevant resources

Have A Test Plan



- Include a repeatable way to test/run your change
- Everyone loves demos, especially if they're visual (videos/images)
 - Front-end: UI/UX
 - Back-end: Logs, full-stack, Postman + JSON
- Basic trust token to show that your code works
- **Every pull request is a piece of product history**



No Test Plan

- *"What exactly does this do?"*

- *"Are you sure this actually works?"* 
- (Future teammates) *"How does this flow work again?"*




Thorough Test Plan

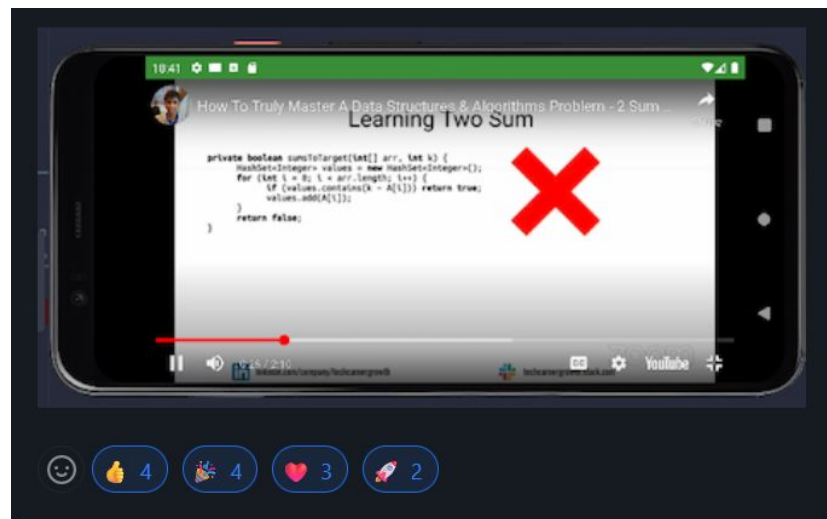
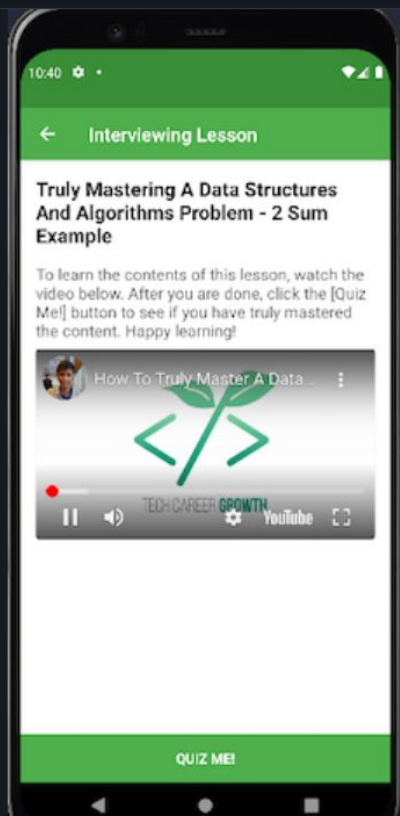
- *"It's so cool seeing exactly what this adds!"* 
- *"Oh wow - This code really is battle tested."* 
- (Future teammates) *"Regression testing this is a breeze!"* 

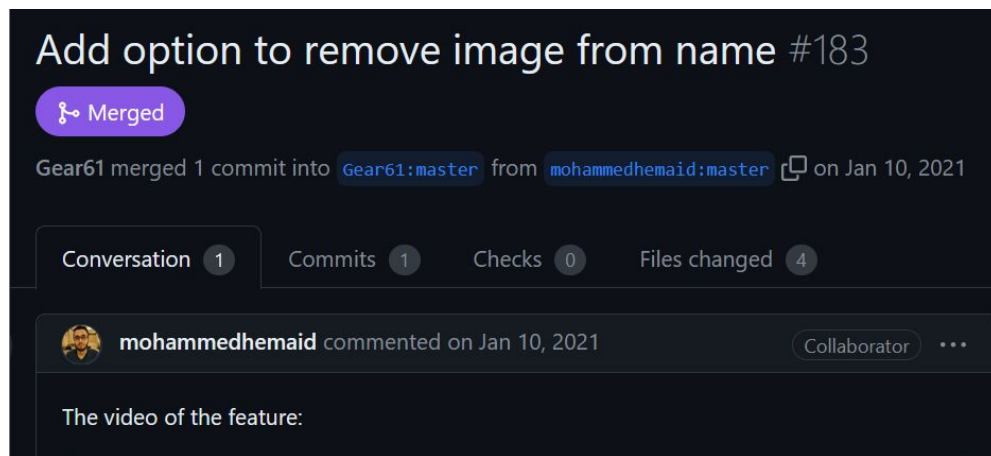
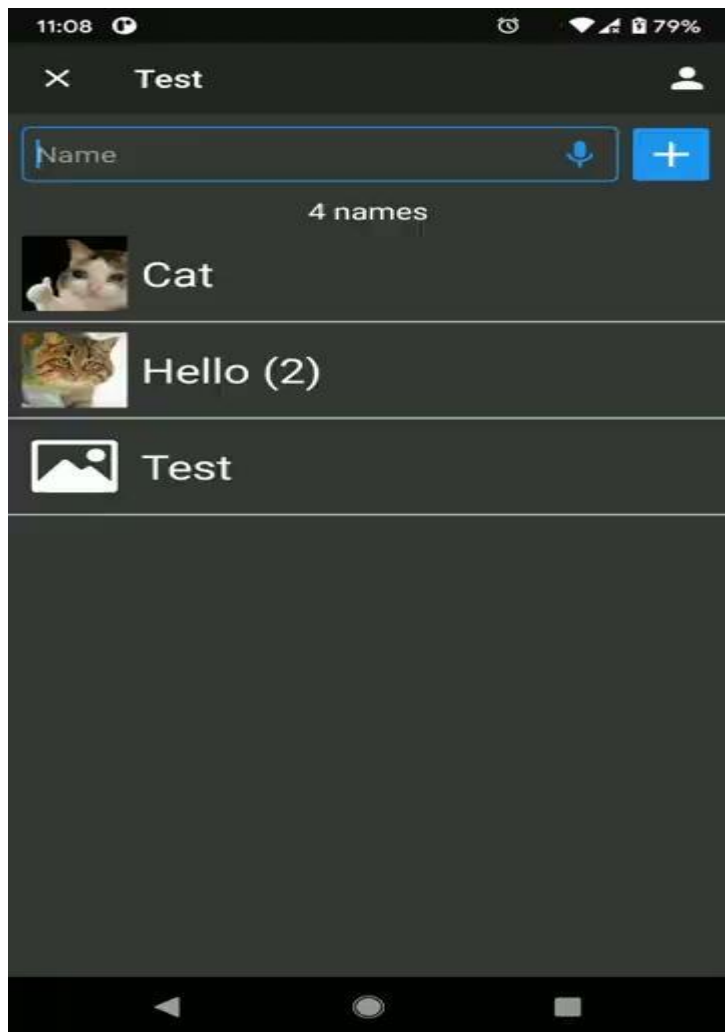


 Closed

Feature/youtube-full-screen #19

prnvptl wants to merge 4 commits into [Gear61:master](#) from ...





Code Review Feedback



- **Code review feedback is one of the greatest gifts you can possibly receive**
- Core ingredient to ramping up technical strength
- Make sure to:
 - Respond graciously with an appreciative tone
 - Respond quickly
 - Understand the core principles behind the feedback



How To Grow Lightning Fast As An Engineer ⚡

1. Publish clean pull requests that are easy to comment on 📄
2. Get tons of incredible feedback from teammates 📦
3. Treat that feedback as a gift and incorporate it 🤝
4. Never make the same mistakes that feedback covers ever again 🎯
5. Repeat Steps #1 - #4 until you're a clean code publishing machine respected by your entire team 💪

DO NOT Be This Person 🙄

**Ignores feedback
and tries to land
anyways**

*"That's such a small
issue - I don't think it's
worth addressing."*

*"I need to move
fast on this - Can
we do this later?"*

*"But this is how I
have always done it
before."*

*"That's such a
narrow case - It won't
happen in prod."*

*"I'll address it in the
next pull request."*

"I'm happy to do that (I'm fine either way), but I had it on the same line as the conditional **since that's how it is on other major parts of the codebase.**

Zooming out, it seems like we need a consistent style guide across the team. Want to work together on that and try to incorporate it into a linter?"

"Can you move the bracket to after the if statement?"

Common Messy Code Patterns

Code Comment “Caches”

```
93
94  getKeywords(QUERY)
95      .then(async results => {
96          let { keywords } = results
97
98          // Expand on the first EXPAND_AMT keywords
99          .....// let kwToExpand = keywords.slice(1, EXPAND_AMT+1)
100         .....// for(let i=0; i<kwToExpand.length; i++){
101         .....// .....let kw = kwToExpand[i]
102         .....// .....let extraKws = await getKeywords(kw)
103         .....// .....extraKws.forEach(ekw => {
104         .....// .....    if(!keywords.includes(ekw))
105         .....// .....        keywords.push(ekw)
106         .....// .....})
107         .....// .....}
108
109         fs.writeFileSync(`keywords/${FILE_NAME}.json`, JSON.stringify
110         console.log('Keywords Generated: '+keywords.length)
111     })
```

- This is usually done when a feature is getting removed, but it may come back someday
- **Idea:** When that day comes, we just comment the code back in
- **Reality:** The day never comes and the huge comment block makes the file hard to read
- **Better option:** Just delete the code and learn how to use version control



git

"Don't be lazy
- Learn how I
work!"

Add back legacy home feed small video card

[Browse files](#)

master

 Gear61 committed on Jul 13, 2023

1 parent 76534ec commit 1b97f07

Showing 2 changed files with 58 additions
and 0 deletions.

[Whitespace](#)[Ignore whitespace](#)[Split](#)[Unified](#)

48

app/src/main/res/layout/coach_video_reference_item.xml

@@ -0,0 +1,48 @@

```
1 + <?xml version="1.0" encoding="utf-8"?>
2 + <androidx.cardview.widget.CardView
3 +     xmlns:android="http://schemas.android.com/apk/res/
        android"
4 +     xmlns:tools="http://schemas.android.com/tools"
5 +     android:layout_width="256dp"
6 +     android:layout_height="wrap_content"
7 +     android:layout_marginHorizontal="9dp"
```

Remember, the key is to provide feedback that helps improve the code quality while maintaining a positive and supportive environment.

Check out the related resources below to learn more 🖱️

Code Review Best Practices



[Masterclass] How To Do Amazing Code Review



"Advice for getting reviews?"



What's on your mind?



Non-Descriptive Names

```
// Avoid Single Letter Names
```

```
let n = 'use name instead'
```

```
// Avoid Acronyms
```

```
let cra = 'no clue what this is'
```

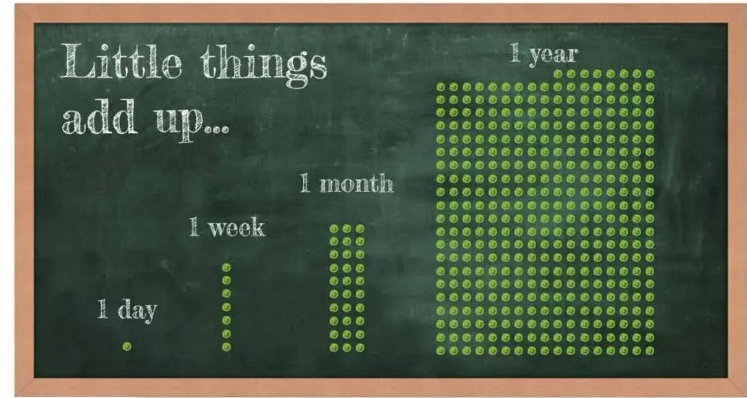
```
// Avoid Abbreviations
```

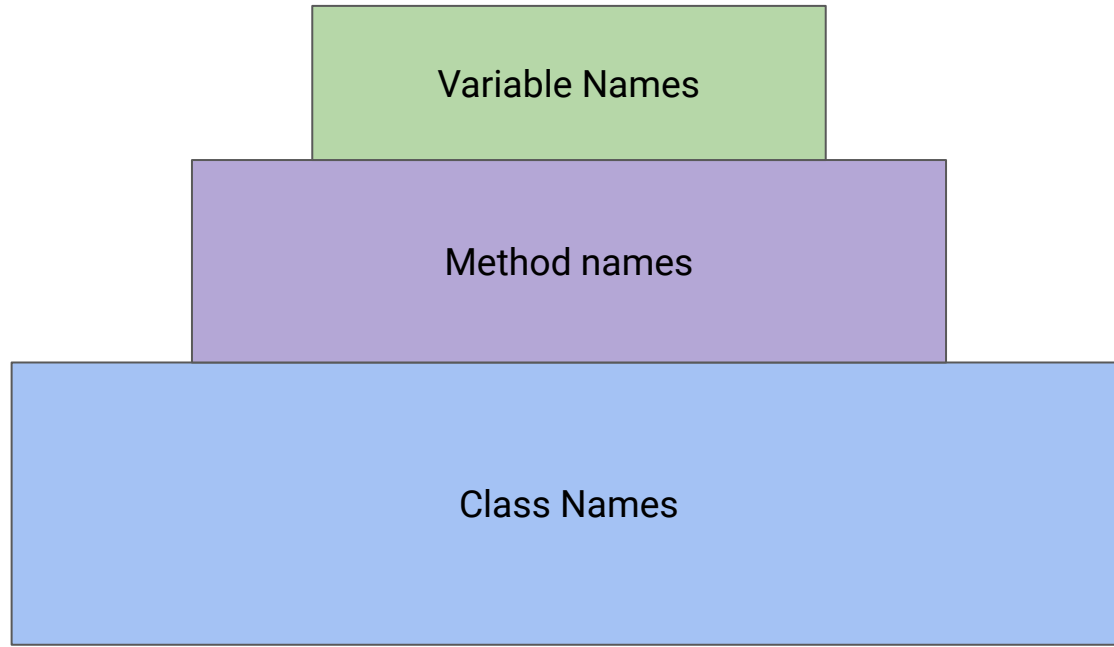
```
let cat = 'cat or category??'
```

```
// Avoid Meaningless Names
```

```
let foo = 'what is foo??'
```

- A lot of school and tutorial materials have lazy names for variables, classes, methods
- **Idea:** Naming is hard, and I will save time by being lazy. Also, efficiency is cool
- **Reality:** The time save is eradicated as people struggle to read the code
- **Better option:** Take <30 seconds to come up with good, descriptive names





Bigger block size = Spend more time 🕒

```

fun capitalizeSentences(
    locale: Locale,
    input: String
): String {
    val instance = BreakIterator.getSentenceInstance(locale)
    instance.setText(input)

    var a = instance.first()
    var b = instance.next()
    val strB = StringBuilder()
    while (b != BreakIterator.DONE) {
        val subS = input.substring(a, b)
        strB.append(subS.replaceFirstChar { if (it.isLowerCas
        a = b
        b = instance.next()
    }
    return strB.toString()
}

```



```

fun capitalizeSentences(
    locale: Locale,
    input: String
): String {
    val sentenceIterator = BreakIterator.getSentenceInstance(locale)
    sentenceIterator.setText(input)

    var startIndex = sentenceIterator.first()
    var endIndex = sentenceIterator.next()
    val output = StringBuilder()
    while (endIndex != BreakIterator.DONE) {
        val extractedSentence = input.substring(startIndex, endIndex)
        output.append(extractedSentence.replaceFirstChar { if (it.isLo

        // Advance the indices to capture the next sentence
        startIndex = endIndex
        endIndex = sentenceIterator.next()
    }
    return output.toString()
}

```

- Think about the code flow in English, and translate it over 📝✍️
- Assume everyone reading your codebase is an onboarding new-grad 🧐

Too Many Comments



- This is another thing schools will do to make extra sure students understand
- **Idea:** With all the comments ever, my code will be the easiest to understand
- **Reality:** Code is physically way longer and a slog to read
- **Better option:** Write better code that is self-explanatory and just leave out the comments



- Comments just describe what the code is clearly already doing 😐
- Level up the readability by having better variable names instead 🧠

```
main.py x
1      # initializing two variables
2      a = 5
3      b = 7
4
5      # adding a and b
6      result = a + b
7
8      # printing the result of the addition
9      print(result)
10     |
```



```
fun printSum(firstNumber: Int, secondNumber: Int) {
    val sum = firstNumber + secondNumber
    println(sum)
}
```

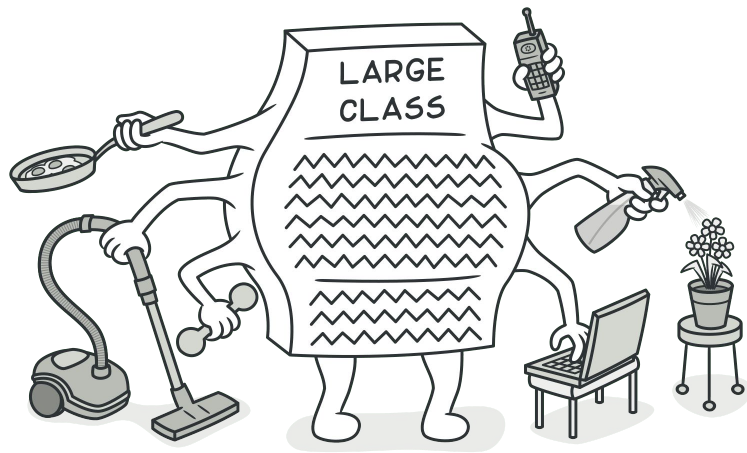
Everything In 1 Place

```
;; The new window is the minimap
(setq buffer-window (selected-window))
(select-window (split-window-horizontally
  (* -1 width)))

;; Set up the minimap window:
;; You should not be able to enter the minimap window.
(set-window-parameter nil 'no-other-window t)
;; Switch to buffer.
(switch-to-buffer
  (get-buffer-create minimap-buffer-name) t t)
;; Do not fold lines in the minimap.
(setq truncate-lines nil)
;; Make it dedicated.
(when minimap-dedicated-window
  (set-window-dedicated-p nil t))
;; Return minimap window, but make sure we select the window where
;; the buffer is in.
(progn
  (selected-window)
  (select-window buffer-window)))

(defun minimap-setup-hooks (&optional remove)
  "Hook minimap into other modes.
If REMOVE is non-nil, remove minimap from other modes."
  (if remove
    (progn
      (remove-hook 'outline-view-change-hook 'minimap-sync-overlays)
      (remove-hook 'hs-hide-hook 'minimap-sync-overlays)
      (remove-hook 'hs-show-hook 'minimap-sync-overlays)
      (remove-hook 'flycheck-after-syntax-check-hook 'minimap-sync-overlays))
    ;; outline-(minor-)mode
    (add-hook 'outline-view-change-hook 'minimap-sync-overlays)
    ;; hide/show
    (add-hook 'hs-hide-hook 'minimap-sync-overlays)
    (add-hook 'hs-show-hook 'minimap-sync-overlays)
    (add-hook 'flycheck-after-syntax-check-hook 'minimap-sync-overlays)))
```

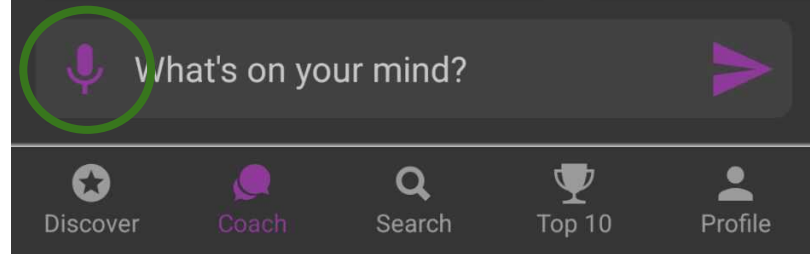
- Also referred to as “god classes”
- It’s easy to put lots of code in just 1 file or module
- **Idea:** Everything I care about is in 1 place and I don’t need to worry about dependencies
- **Reality:** Code is unreadable and builds slowly
- **Better option:** Learn interfaces and other modularization techniques for <1k LoC files



```
package com.randomappsinc.techcareergrowth.coach

import ...

/** Primary view controller for TaroGPT */
class CoachFragment: Fragment(), CoachChatAdapter.Listener,
```



```
if (conversationManager.isCareerCoachThinking()) {
    UIUtil.showLongToast(
        stringId = "Please wait for your TaroGPT to respond before sharing ...",
        context = requireContext()
    )
    return
}

val speechRecognizer = SpeechRecognizer.createSpeechRecognizer(activity)
speechRecognizer!!.setRecognitionListener(this)
val speechRecognizerIntent = Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH)
speechRecognizer.startListening(speechRecognizerIntent)
val speechToTextDialog = SpeechToTextDialog(
    activity = requireActivity(),
    listener = this
)
speechToTextDialog.maybeShowDialog()
```



```
if (conversationManager.isCareerCoachThinking()) {
    UIUtil.showLongToast(
        stringId = "Please wait for your TaroGPT to respond before sharing ...",
        context = requireContext()
    )
    return
}
speechToTextManager.startSpeechToTextFlow()
```



```

class SpeechToTextManager(
    private var activity: Activity?,
    private val listener: Listener
) : RecognitionListener, SpeechToTextDialog.Listener {

    interface Listener {
        fun onTextSpoken(spokenText: String)
    }

    // This is lazily instantiated and is also nulled out when the user dismisses the prompt without speaking
    private var speechRecognizer: SpeechRecognizer? = null
    private val speechRecognizerIntent = Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH)
    private val speechToTextDialog: SpeechToTextDialog

    init {
        speechRecognizerIntent.putExtra(
            RecognizerIntent.EXTRA_LANGUAGE_MODEL, RecognizerIntent.LANGUAGE_MODEL_FREE_FORM
        )
        speechRecognizerIntent.putExtra(
            RecognizerIntent.EXTRA_LANGUAGE,
            Locale.getDefault().language
        )
        speechRecognizerIntent.putExtra(RecognizerIntent.EXTRA_PARTIAL_RESULTS, value: true)
        speechToTextDialog = SpeechToTextDialog(
            activity = activity!!,
            listener = this
        )
    }
}

```

```

/** Primary view controller for TaroGPT */
class CoachFragment: Fragment(), CoachChatAdapter.Listener,
    SentiyenRestClient.Listener, SpeechToTextManager.Listener {

```

```

override fun onTextSpoken(spokenText: String) {
    var finalText = spokenText
    if (!spokenText.endsWith(suffix: "?")) {
        finalText += "?"
    }
    binding.messageInput.setText(finalText)
}

```

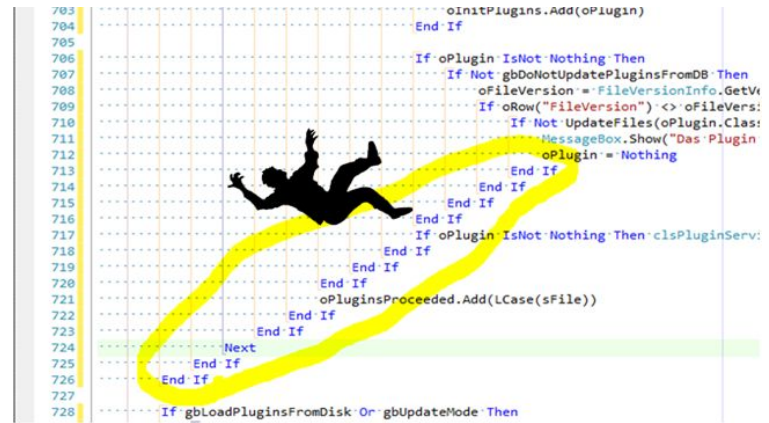
Break down systems into components, and figure how they should talk to each other. 💡

Infinite Layers Of Nesting

```
function register()
{
    if (!empty($_POST)) {
        $msg = '';
        if ($_POST['user_name']) {
            if ($_POST['user_password_new']) {
                if ($_POST['user_password_new'] === $_POST['user_password_repeat']) {
                    if (strlen($_POST['user_password_new']) > 5) {
                        if (strlen($_POST['user_name']) < 65 && strlen($_POST['user_name']) > 1) {
                            if (preg_match('/^[a-z\d]{2,64}$/i', $_POST['user_name'])) {
                                $user = read_user($_POST['user_name']);
                                if (!isset($user['user_name'])) {
                                    if ($_POST['user_email']) {
                                        if (strlen($_POST['user_email']) < 65) {
                                            if (filter_var($_POST['user_email'], FILTER_VALIDATE_EMAIL)) {
                                                create_user();
                                                $_SESSION['msg'] = 'You are now registered so please login';
                                                header('Location: ' . $_SERVER['PHP_SELF']);
                                                exit();
                                            } else $msg = 'You must provide a valid email address';
                                        } else $msg = 'Email must be less than 64 characters';
                                    } else $msg = 'Email cannot be empty';
                                } else $msg = 'Username already exists';
                            } else $msg = 'Username must be only a-z, A-Z, 0-9';
                        } else $msg = 'Username must be between 2 and 64 characters';
                    } else $msg = 'Password must be at least 6 characters';
                } else $msg = 'Passwords do not match';
            } else $msg = 'Empty Password';
        } else $msg = 'Empty Username';
        $_SESSION['msg'] = $msg;
    }
    return register_form();
}
```



- Control flows are naturally hard to follow
- **Idea:** Put all my if-else logic in 1 place, so I don't mess it up and can be lazy
- **Reality:** Code physically runs off the screen and sucks to read
- **Better option:** Move code into other methods and use tactics like early-return



```
703 oInitPlugins.Add(oPlugin)
704 End If
705
706 If oPlugin IsNot Nothing Then
707     If Not gbDoNotUpdatePluginsFromDB Then
708         oFileVersion = FileVersionInfo.GetVersionInfo(oPlugin.FileName).FileVersion
709         If oRow("FileVersion") <> oFileVersion Then
710             If Not UpdateFiles(oPlugin.ClassName, oPlugin.FileName) Then
711                 MsgBox.Show("Das Plugin " & oPlugin.Name & " konnte nicht aktualisiert werden.")
712             End If
713         End If
714     End If
715 End If
716 End If
717 If oPlugin IsNot Nothing Then clsPluginServ = New clsPluginService(oPlugin)
718 End If
719 End If
720 End If
721 oPluginsProceeded.Add(LCase(sFile))
722 End If
723 End If
724 Next
725 End If
726 End If
727
728 If gbLoadPluginsFromDisk Or gbUpdateMode Then
```

Early Return

```
override fun onDiscoverFeedsFetched() {  
    if (isAdded) {  
        requireActivity().runOnUiThread {  
            homeFeedAdapter.onDiscoverFeedsFetched()  
        }  
    }  
}
```



```
override fun onDiscoverFeedsFetched() {  
    if (!isAdded) {  
        return  
    }  
    requireActivity().runOnUiThread {  
        homeFeedAdapter.onDiscoverFeedsFetched()  
    }  
}
```

Merging Conditionals

```
if (data != null) {  
    if (data.isNotEmpty()) {  
        val rawTranscription = data[data.size - 1]  
        val latestPartialTranscription = StringUtil.capitalizeSentences(  
            locale = activity!!.resources.configuration.locales.get(0),  
            input = rawTranscription  
        )  
        speechToTextDialog.setMessage(message = latestPartialTranscription)  
    }  
}
```



```
if (data != null && data.isNotEmpty()) {  
    val rawTranscription = data[data.size - 1]  
    val latestPartialTranscription = StringUtil.capitalizeSentences(  
        locale = activity!!.resources.configuration.locales.get(0),  
        input = rawTranscription  
    )  
    speechToTextDialog.setMessage(message = latestPartialTranscription)  
}
```

```
/**
 * Use this to open any sort of content.
 * Has mechanisms to fetch a fresh version of content if applicable.
 */
class ContentOpener(var activity: Activity?) {
```

```
fun openLesson(
    lesson: Lesson,
    lessonList: ArrayList<Lesson>? = null
) {
    if (lesson.firebaseId.isBlank()) {
        if (activity != null) {
            UIUtil.showLongToast("Failed to load
        }
        return
    }
}
```



```
fun openLesson(
    lesson: Lesson,
    lessonList: ArrayList<Lesson>? = null
) {
    if (lesson.firebaseId.isBlank() && activity != null) {
        UIUtil.showLongToast("Failed to load career advice.
    }
    return
}
```

Be careful! Double check
all control flows. ✓

Seas Of Newlines

```
VersionRequirementManager(activity = this).startVerifying()
```

```
contentOpener = ContentOpener(activity = this)
```

```
navigationController = HomepageFragmentController(supportFragmentManager, R.id.container)
```

```
binding.bottomNavigation.setOnClickListener(this)
```

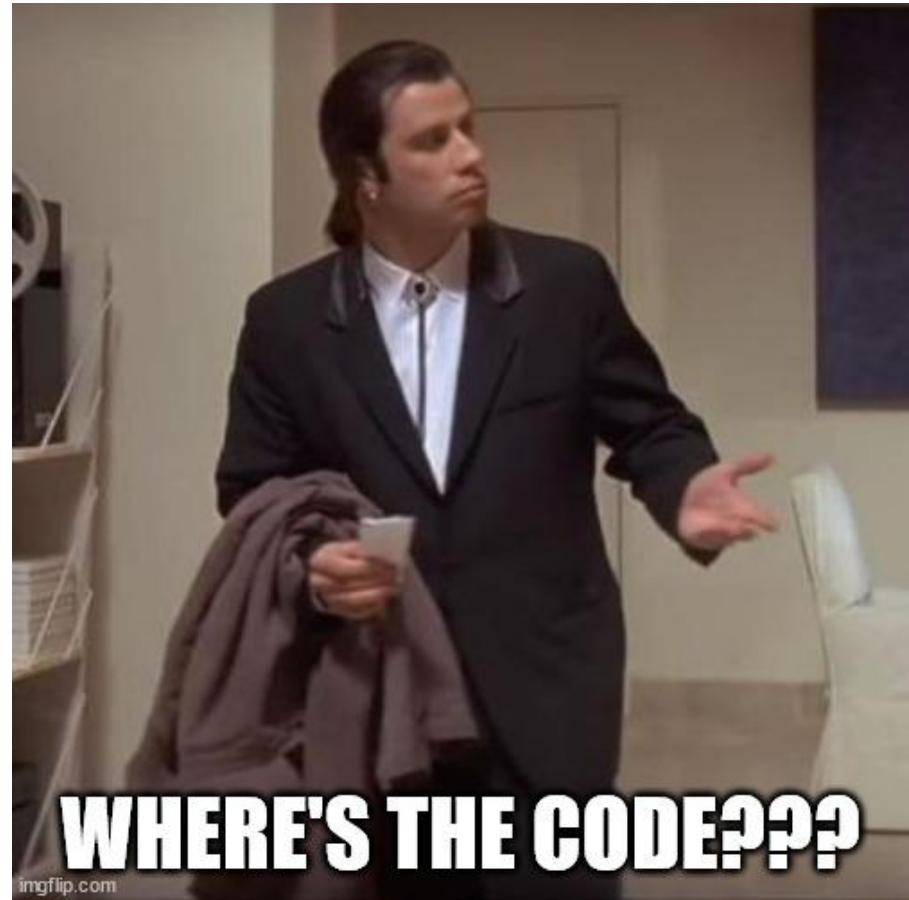
```
if (savedInstanceState == null) {
```

```
    binding.bottomNavigation.setCurrentSelected(R.id.coach_button)
```

```
val lessonTag = intent.getStringExtra(LESSON_TAG_KEY)
```

```
val lessonId = intent.getStringExtra(LESSON_ID_KEY)
```


- This one is confusing but weirdly common
- **Idea:** The newlines will cleanly separate different code block types from one another
- **Reality:** Code is just way longer and sucks to read
- **Better option:** Move code into other methods and classes to cleanly modularize



Wrapping Up

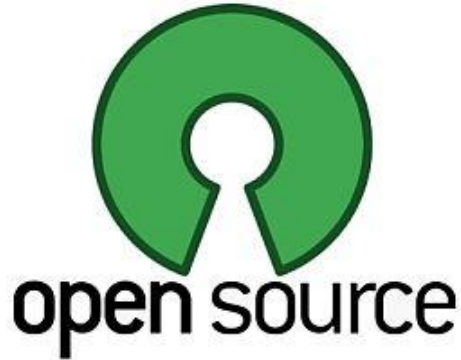
Level Up Even Faster

GOTTA GO

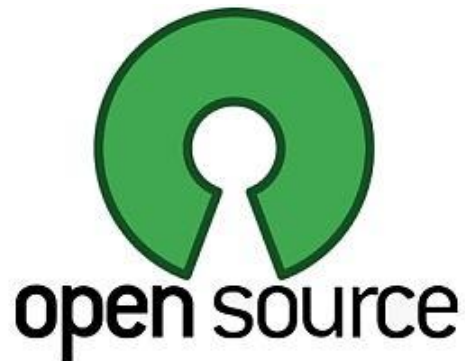


FAST

~~“There’s no way I can learn all this
code quality stuff if I don’t work at a
fancy tech company like FAANG.”~~



Remember: There's No Shortcuts



- Don't just release v1 and run away
 - Respond to user feedback (even the angry ones)
 - If your code can't scale, improve it
- Don't just lazily tackle the easiest issues
 - Try to actually understand the repo
 - Find a repo you use and care about



... build incredible software! 🧑💻 🚀 💪

Make your pull requests a joy to review.

Do the right thing. Do the hard things.

Become world-class at your craft.

The end-user comes first.

Just build.

What next?

- Share + get support: joinTaro.com/questions/create/
 -  Get 10x better advice than anything on Blind or Reddit
- Find community: joinTaro.com/events/
 -  Join me for Group Office Hours!

thank you 

team+codequality@jointaro.com